

Online Nash Social Welfare Maximization with Byzantine-Resilient Trust Mechanisms and Graph-Clustering-Based Detection

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Abstract—In this work, we study the online Nash Social Welfare (NSW) allocation problem in the presence of an unknown subset of malicious or faulty agents. In contrast to prior works that assume honest reporting, we generalize and characterize NSW dynamics under binary sequences of malicious transmissions with dynamic, time-varying attack rates, using trust observations derived from physical side-channels. We adapt the Trusted Neighborhood Learning algorithm to the NSW setting, where trust observations drive a *hybrid allocation rule* that combines an equal-share fairness floor with a greedy softmax component grounded in the *predicted utility under promised share* (Definition 1). Under these conditions, legitimate agents almost surely classify their neighbors and achieve consensus on resource allocation with near-geometrically decaying misclassification probabilities.

We introduce three new attack types that substantially narrow the detection difficulty gap: TRUST-MIMICRY, in which an attacker tracks the running threshold ξ_t to stay perpetually inside the trusted zone; REPUTATION-POISON, in which an agent accumulates high trust during a warm-up window then switches to full aggression; and DRIFTING, in which the malicious trust mean $E_{\mathcal{M}}(t)$ rises toward $E_{\mathcal{L}}$ over time, violating the stationarity assumption underlying existing detection guarantees. We further integrate the sparsest-subgraph graph clustering approach of Moradi et al. [23] as a third detection benchmark, operating on a blended affinity matrix $W_{ij} = \sigma \cdot T_{ij} + (1 - \sigma) \cdot S_{ij}$ that combines normalised trust T_{ij} with Pearson valuation-similarity S_{ij} . Across all ten attack types, the sparsest-subgraph detector achieves F1 = 1.0 on TRUST-MIMICRY attacks where Algorithm 1 reaches only 8% detection rate. Domain-specific scenarios (IoT sensor networks, spectrum sharing, distributed compute clusters, adversarial-overlap stress tests) and a fairness-efficiency analysis via the allocation blend parameter α complete the evaluation. Numerical experiments with 200 rounds and 20 independent trials validate convergence, demonstrate bounded deviation, and show that the calibrated trust gap ($E_{\mathcal{L}} - E_{\mathcal{M}} = 0.11$, $\xi_0 = 0.25$) produces distinct asymptotic detection slopes per attack type—a regime absent from prior work with trivially large gaps.

Index Terms—Nash Social Welfare, Byzantine resilience, trust mechanisms, multi-agent systems, online allocation, malicious agent detection, graph clustering, sparsest subgraph, hybrid allocation.

I. INTRODUCTION

IN resource allocation problems, a set of agents aim to achieve fair division using local computations and interactions over a communication network. The Nash Social Welfare (NSW) problem serves as a foundation for fair division [1], distributed optimization [2], and multi-agent coordination [3]. NSW algorithms relying on the full cooperativeness of each

agent are vulnerable to malicious and faulty inputs from strategic participants [4], [5].

In this paper, we analyze the NSW allocation problem in the presence of malicious agents whose attack behavior may evolve over time. We consider dynamic attack rates, where malicious agents can make attack decisions based on their own history, resulting in potentially dependent and strategic sequences of malicious behavior. The premise of this paper is to address and mitigate such dynamic malicious behavior based on detection using quantifiable “trust” observations as side information derived from the physical aspects of the communication network.

A critical gap in existing work is that reported experiments use trust parameters with gap $E_{\mathcal{L}} - E_{\mathcal{M}}$ so large that detection occurs within the first round, making convergence plots trivial and obscuring the qualitative differences between attack types. We derive and apply calibrated parameters ($E_{\mathcal{L}} = 0.63$, $E_{\mathcal{M}} = 0.52$, $\xi_0 = 0.25$, $p = 0.55$) under which the expected cumulative gap grows at rate 0.060 per round while the threshold $\xi_t \sim \xi_0 \sqrt{t \ln t}$ grows sublinearly—producing detection times that range from 30 rounds (CONSTANT-HIGH) to never-detected (TRUST-MIMICRY, DRIFTING) within 200 rounds. This calibration is the necessary foundation for observing asymptotic behavior and comparing detector performance meaningfully.

A. Motivation

Fair resource allocation is critical in multi-agent systems across diverse domains.

Cloud computing: Platforms must allocate CPU time, memory, and bandwidth among users. Fraudulent users may over-report needs to monopolize resources, degrading service quality for legitimate users.

Wireless networks: Base stations allocate bandwidth to devices. Compromised devices may manipulate demand reports to disrupt fair allocation and degrade network performance.

Energy grids: Demand management systems coordinate electricity distribution. Strategic actors may manipulate reports to influence prices or gain unfair advantage during peak periods.

Decentralized finance: Asset pooling mechanisms distribute resources among participants. Strategic misreporting can lead to unfair share allocation and system instability.

IoT sensor networks: Low-power sensors with sparse connectivity report environmental readings. Intermittently-attacking compromised sensors are harder to detect than persistent attackers and cause disproportionate harm to consensus quality [20], [21].

Traditional NSW algorithms degrade significantly under such manipulation. Our experiments show NSW ratio degradation of up to 15% without trust-based detection under the calibrated trust gap, and up to 18% when adversary count grows to $|\mathcal{M}| = 6$, significantly higher than previously reported because the calibrated gap forces the system to operate in a genuinely challenging detection regime.

B. Related Work

Resilient consensus methods dealing with malicious agents have different methodologies. Some studies utilize only transmitted data for detection [5], [8]–[10], but have restrictions on network connectivity and bounded number of malicious agents. Since these restrictions affect solving related multi-agent problems such as optimization [10] and spectrum sensing [11], another body of literature proposes using additional side information for assessment of agent identity [12]–[15].

The work [12] details computation of stochastic trust observations $\alpha_{ij}(t) \in [0, 1]$ from wireless signal information. The work [16] proves that malicious agents can be detected via trust observations and trustworthy agents can reach consensus even when malicious agents constitute the majority. However, [16] employs fixed threshold detection which may fail when malicious agents exhibit dynamic or random behavior.

Graph-based clustering approaches for trust-aware systems have been studied in the recommendation system literature [23], where user–item similarity graphs are clustered to improve recommendation quality under sparse data and cold-start conditions. Moradi et al. propose building affinity graphs from trust statements blended with rating similarity, and finding the *sparsest subgraph*—nodes with minimum internal density—as automatically-identified cluster seeds. We adapt this machinery to the multi-agent detection problem: the β_{ij} trust matrix is our affinity graph, and the sparsest-subgraph algorithm identifies the outlier (malicious) cluster without requiring a pre-specified cluster count or a threshold parameter.

No prior work bridges Byzantine-resilient consensus with NSW allocation while simultaneously maintaining preference heterogeneity, providing a calibrated trust gap analysis, and benchmarking graph-clustering-based detection against the threshold-based detector.

C. Contributions

Our contributions are summarized as follows:

1) *Novel Integration:* The first integration of trust-based detection from distributed consensus into online NSW allocation, properly adapted to respect preference heterogeneity through independent side-channel observations.

2) *Hybrid Allocation Rule (α -split):* We replace the pure greedy softmax with a convex combination $z_i = \alpha \cdot (B/N) + (1 - \alpha) \cdot z_i^{\text{greedy}}$, where the equal-share component $\alpha \cdot B/N$ provides a fairness floor and the greedy component $(1 - \alpha) \cdot$

z_i^{greedy} drives NSW efficiency. The parameter $\alpha \in [0, 1]$ traces the efficiency–fairness Pareto frontier (Fig. 8).

3) *Three New Attack Types:* TRUST-MIMICRY tracks ξ_t to stay inside the detection boundary; REPUTATION-POISON accumulates trust during a warm-up window then attacks; DRIFTING gradually erodes the trust gap $E_{\mathcal{L}} - E_{\mathcal{M}}$ over time. Together with the existing seven types, ten attack strategies are evaluated.

4) *Calibrated Trust Gap Analysis:* We derive analytically and validate empirically that previous parameter choices (gap = 0.63, $\xi_0 = 0.15$) make detection trivial within one round. Calibrated parameters (gap = 0.11, $\xi_0 = 0.25$, $p = 0.55$) place the system in the genuinely challenging detection regime, producing distinct asymptotic slopes across attack types.

5) *Moradi Graph-Clustering Detection Benchmark:* We introduce a third detector based on the sparsest-subgraph algorithm of [23]. The blended affinity matrix $W_{ij} = \sigma T_{ij} + (1 - \sigma) S_{ij}$ combines normalised cumulative trust T_{ij} with Pearson valuation-similarity S_{ij} . The sparsest-subgraph detector achieves F1 = 1.0 on TRUST-MIMICRY where Algorithm 1 reaches only 8% and spectral clustering reaches 0.47.

6) *Moradi Cluster Merging Safety Net:* A merging step absorbed from [23] prevents starvation of legitimate agents whose trusted clusters fall below a minimum size—directly improving the min-agent utility metric under adversarial conditions.

7) *Domain Scenario Library:* Named presets (IoT networks, spectrum sharing, distributed compute, adversarial overlap) provide realistic valuation distributions (log-normal, Pareto), sparse connectivity, and domain-tuned parameters, grounding every experiment in an interpretable real-world context.

8) *Asymptotic Convergence and Deviation:* Relying on almost-sure correct classification, we show legitimate agents reach NSW consensus almost surely (Corollary 1). For a given attack probability, the maximal deviation from nominal NSW is characterized (Theorem 1), and geometric convergence rate is established (Theorem 2).

II. NSW DYNAMICS WITH MALICIOUS AGENTS

A. Notation

We denote the absolute value of a scalar and cardinality of a finite set by $|\cdot|$. We write x_i and A_{ij} for the i -th entry of vector x and ij -th entry of matrix A , respectively. The backward matrix product is:

$$\prod_{k=\tau}^t H(k) := \begin{cases} H(t) \cdots H(\tau+1)H(\tau) & \text{if } t \geq \tau, \\ I & \text{otherwise,} \end{cases} \quad (1)$$

where I is the identity matrix.

B. NSW in Presence of Malicious Agents

We consider the NSW allocation process among agents defined by set $\mathcal{N} = \{1, \dots, N\}$. Agents exchange information through static undirected graph $G(\mathcal{N}, E)$. The agent set consists of disjoint legitimate agents $\mathcal{L}$ and malicious agents $\mathcal{M}$, satisfying $\mathcal{L} \cup \mathcal{M} = \mathcal{N}$, $\mathcal{L} \cap \mathcal{M} = \emptyset$.

Valuations: Each agent i has true valuation $v_i^{\text{true}}(t)$, drawn from domain-specific distributions (log-normal with $\sigma = 0.80$

TABLE I
ATTACK-TYPE-SPECIFIC DISTORTION INTERVALS $[\varphi_{\min}, \varphi_{\max}]$ AND
EFFECTIVE ATTACK PROBABILITIES.

Attack type	$\varphi_{\min}$	$\varphi_{\max}$	p_m profile
Constant high	0.05	8.0	Fixed 0.55
Constant low	0.60	1.40	Fixed ≈ 0.15
Intermittent	0.10	5.0	Sinusoidal, peak $t = 60$
Strategic burst	0.02	10.0	0.92 in burst, 0.12 else
Adaptive (undetected)	0.02	12.0	Up to 0.97
Adaptive (detected)	0.85	1.15	≈ 0.10
Coordinated (burst)	0.02	9.0	0.95 in burst
Random	0.10	6.0	$U(0.10, 0.95)$
Trust-Mimicry	0.50	2.00	0.22 when safe, 0.03 else
Reputation-Poison	0.95	1.05	≈ 0.02 for $t < K_w$, then 0.98
Drifting	0.15	4.0	Phase-modulated; $E_{\mathcal{M}}(t) \nearrow$

for compute/IoT, Pareto with shape $a = 2$ for spectrum sharing, or Dirichlet). Legitimate agents report truthfully; malicious agents distort: $v_m^{\text{report}}(t) = \varphi_m(t) \cdot v_m^{\text{true}}(t)$.

Utilities and NSW:

$$u_i = \sum_{t=1}^T v_i^{\text{true}}(t) \cdot z_i(t), \quad \text{NSW}_{\mathcal{L}} = \exp\left(\frac{1}{|\mathcal{L}|} \sum_{i \in \mathcal{L}} \ln u_i\right). \quad (2)$$

Allocation dynamics: Legitimate agent values update as

$$x_i(t+1) = w_{ii}(t)x_i(t) + \sum_{j \in \mathcal{N}_i} w_{ij}(t)x_j(t), \quad (3)$$

with weights defined by the hybrid allocation rule (Section II-E). In matrix form:

$$x_{\mathcal{L}}(t+1) = [W_{\mathcal{L}}(t) \quad W_{\mathcal{M}}(t)] \begin{bmatrix} x_{\mathcal{L}}(t) \\ x_{\mathcal{M}}(t) \end{bmatrix}. \quad (4)$$

For all $t \geq T_0$, we decompose $x_{\mathcal{L}}(T_0, t) = \tilde{x}_{\mathcal{L}}(T_0, t) + \phi_{\mathcal{M}}(T_0, t)$, where $\tilde{x}_{\mathcal{L}}$ results from legitimate dynamics and $\phi_{\mathcal{M}}$ captures malicious influence.

C. Attack and Trust Models

For malicious agent $m \in \mathcal{M}$, the attack decision at time t is $f_m(t) \in \{0, 1\}$. When attacking, the distortion factor $\varphi_m(t)$ is drawn from an attack-type-specific interval. Table I summarizes all ten attack strategies.

Coordinated attack. The coalition $\mathcal{M}$ synchronises every K rounds. In burst round t ($t \bmod K = 0$):

$$v_m^{\text{report}}(t) = A \cdot v_m^{\text{true}}(t) \cdot U_m, \quad m \in \mathcal{M}, \quad (5)$$

$$v_{i^*}^{\text{report}}(t) = \varphi^* \cdot v_{i^*}^{\text{true}}(t), \quad (6)$$

where $U_m \sim U(0.8, 1.2)$, $A = 9$, $\varphi^* \sim U(0.01, 0.08)$, and $i^*(t) \in \mathcal{L}$ is a rotating legitimate target.

Trust-Mimicry attack. The attacker observes the current threshold ξ_t and its own average beta gap $\Delta\beta_m(t)$ (as seen by legitimate observers). It attacks only when $\Delta\beta_m(t) < \xi_t - \varepsilon_{\text{margin}}$ with $\varepsilon_{\text{margin}} = 0.02$, keeping its accumulated trust perpetually inside the trusted zone. Distortion is mild ($\varphi \sim U(0.50, 2.00)$) to avoid detection through report anomalies.

Reputation-Poison attack. The agent behaves honestly for a warm-up window of $K_w = 40$ rounds, accumulating high β_{im} scores. After warm-up, it switches to full aggression

($p_m = 0.98$, $\varphi \sim U(0.02, 9.0)$). Since the cumulative trust accumulator $\beta_{ij}(t) = \sum_{k=0}^t \alpha_{ij}(k)$ has no forgetting, the head-start from honest behavior delays detection even under sustained attack.

Drifting attack. The effective malicious trust mean rises over time:

$$E_{\mathcal{M}}^{\text{eff}}(t) = \min(E_{\mathcal{M}} + r_d \cdot t, E_{\mathcal{L}} - 0.05), \quad (7)$$

with drift rate $r_d = 0.0015$ per round. This violates the stationarity assumption in Assumption 2(1), gradually collapsing the detection margin as t grows.

Trust observations. Each legitimate agent i gathers stochastic trust observation $\alpha_{ij}(t) \in [0, 1]$ from neighbor j , with expectations $E_{\mathcal{L}}$ (legitimate) and $E_{\mathcal{M}}(t)$ (malicious attacking).

Assumption 1 (Legitimate Agents). (1) The subgraph induced by $\mathcal{L}$ is connected. (2) Trust observations $\alpha_{ij}(t)$ for legitimate pairs are independent over time with uniform expected value $E_{\mathcal{L}}$ for all $i \in \mathcal{L}$, $\ell \in \mathcal{N}_i^{\mathcal{L}}$.

Assumption 2 (Malicious Agents). (1) $\mathbb{E}[\alpha_{im}(t)|f_m(t) = 0] = E_{\mathcal{L}}$, $\mathbb{E}[\alpha_{im}(t)|f_m(t) = 1] = \mu_m(t) \leq E_{\mathcal{M}} < E_{\mathcal{L}}$. (2) Given $f_m(t)$, the conditional trust observations are independent over time.

D. Trusted Neighborhood Learning (Algorithm 1)

Each legitimate agent i maintains cumulative trust:

$$\beta_{ij}(t) = \sum_{k=0}^t \alpha_{ij}(k), \quad \forall j \in \mathcal{N}_i. \quad (8)$$

Agent i identifies its most trusted neighbor $\bar{j}(t) \in \arg \max_j \beta_{ij}(t)$ and forms its trusted neighborhood:

$$\hat{\mathcal{N}}_i(t) = \{j \in \mathcal{N}_i \mid \beta_{i\bar{j}}(t) - \beta_{ij}(t) \leq \xi_t\} \cup \{i\}, \quad (9)$$

where $\xi_t = \xi_0 \sqrt{(1+\epsilon)(t+1) \ln(t+2)}$, $\xi_0 = 0.25$, $\epsilon = 0.10$.

Calibration remark. The cumulative gap grows as $\mathbb{E}[\beta_{i\ell}(t) - \beta_{im}(t)] = (E_{\mathcal{L}} - E_{\mathcal{M}}) \sum_k p_m(k) \sim (E_{\mathcal{L}} - E_{\mathcal{M}}) \cdot p \cdot t$ (linear), while $\xi_t \sim \xi_0 \sqrt{t \ln t}$ (sublinear). With the old parameters ($E_{\mathcal{L}} = 0.85$, $E_{\mathcal{M}} = 0.22$, $p = 0.70$, $\xi_0 = 0.15$), the gap rate is 0.441 per round and overtakes ξ_t at $t \approx 1$ —making detection trivially instantaneous regardless of attack type. The calibrated parameters ($E_{\mathcal{L}} = 0.63$, $E_{\mathcal{M}} = 0.52$, $p = 0.55$, $\xi_0 = 0.25$) reduce the gap rate to 0.061 per round, pushing expected detection to 30–200 rounds depending on attack type.

E. Hybrid Allocation Rule (α -split)

Definition 1 (Predicted utility under promised share). For agent i at round t :

$$\tilde{u}_{i,t}(z_1, \dots, z_t) = \frac{\tilde{V}_i}{2N} + z_{i,t}v_{i,t} + \sum_{t'=1}^{t-1} z_{i,t'}v_{i,t'}, \quad (10)$$

where $\tilde{V}_i = \sum_{k=0}^{t-1} v_i^{\text{report}}(k)$ is the running sum of reported valuations.

The priority score for trusted agent i is $P_i(t) = v_{i,t}/\tilde{u}_{i,t}^{\text{base}}$, where $\tilde{u}_{i,t}^{\text{base}}$ is evaluated with $z_{i,t} = 0$. The softmax weights are $w_i = \text{softmax}(\log P_i/\tau)$ with temperature $\tau = 0.30$.

The hybrid allocation combines a fairness floor with merit-based efficiency:

$$z_i(t) = \alpha \cdot \frac{B}{N_{\text{trusted}}} + (1 - \alpha) \cdot w_i \cdot B, \quad i \in \hat{\mathcal{N}}, \quad (11)$$

where $B = 1$ is the per-round budget and $\alpha \in [0, 1]$ is the blend parameter ($\alpha = 0$: pure greedy; $\alpha = 0.5$: half/half; $\alpha = 1$: pure equal split). After allocation, any trusted cluster of size below threshold $m_{\text{merge}} = 2$ is merged with the nearest beta-similar legitimate cluster (Moradi merging step [23]), preventing starvation.

III. GRAPH-CLUSTERING DETECTION: MORADI INTEGRATION

A. Blended Affinity Matrix

Inspired by the trust-similarity blending of Moradi et al. [23], we construct a blended affinity matrix over the trust graph. Define the normalised trust matrix $T_{ij} = \beta_{ij} / \max_k \beta_{ik}$ (symmetrised) and the Pearson valuation-similarity matrix:

$$S_{ij} = \frac{1 + \rho(v_i^{\text{hist}}, v_j^{\text{hist}})}{2} \in [0, 1], \quad (12)$$

where v_i^{hist} is agent i 's valuation history. The blended matrix is:

$$W_{ij} = \sigma \cdot T_{ij} + (1 - \sigma) \cdot S_{ij}, \quad \sigma \in [0, 1]. \quad (13)$$

The parameter σ controls the trust-to-similarity ratio. For TRUST-MIMICRY attacks, the attacker suppresses its β signal but retains anomalous valuation patterns; the S_{ij} component exposes this. Default $\sigma = 0.60$.

B. Sparsest-Subgraph Detection

We implement the approximate sparsest-subgraph algorithm of [23] on W . The density of subgraph $S \subseteq \mathcal{N}$ is:

$$\rho(S) = \frac{\sum_{e \in E(S)} W_e}{|S|}. \quad (14)$$

The algorithm iteratively removes nodes with weighted degree $\geq (2 + 2\varepsilon)\rho(S)$ until a minimum-density residual S^* is found. This residual is the predicted malicious set—no pre-specification of cluster count is required.

Lemma (Moradi [23]). Algorithm 1 of [23] produces a $(2 + 2\varepsilon)$ -approximation to the sparsest subgraph and terminates in $O(\log_{1+\varepsilon} n)$ iterations.

C. Implicit Trust Bootstrap

For agents with fewer than K_b accumulated observations (cold-start case), we supplement trust with an implicit score adapted from Moradi Eq. (5):

$$\tau_{\text{imp}}(i, j) = \frac{|\{t : v_i(t) > \text{med}(v_i) \wedge v_j(t) > \text{med}(v_j)\}|}{|\{t : v_i(t) > \text{med}(v_i)\}|}. \quad (15)$$

Agents whose average implicit trust score falls below 0.35 are temporarily demoted from the trusted set, preventing exploitation of newly-joined agents.

IV. ANALYSIS

A. Misclassification Bounds

Lemma 4. Suppose Assumptions 1 and 2 hold. For legitimate neighbor $\ell \in \mathcal{N}_i^{\mathcal{L}}$:

$$\mathbb{P}(\ell \notin \hat{\mathcal{N}}_i(t)) \leq |\mathcal{N}_i| \cdot e^{-\xi_t^2(t+1)^{-1/2}}. \quad (16)$$

Lemma 5. Suppose Assumptions 1 and 2 hold. For malicious neighbor $m \in \mathcal{N}_i^{\mathcal{M}}$, when $(E_{\mathcal{L}} - E_{\mathcal{M}}) \sum_{k=0}^t p_m(k) - \xi_t > 0$:

$$\mathbb{P}(m \in \hat{\mathcal{N}}_i(t)) \leq e^{-((E_{\mathcal{L}} - E_{\mathcal{M}}) \sum_{k=0}^t p_m(k) - \xi_t)^2(t+1)^{-1/2}}. \quad (17)$$

Remark. Lemma 5 directly explains the difficulty of detecting TRUST-MIMICRY via Algorithm 1: the attacker keeps $\Delta\beta_m(t) < \xi_t$ by construction, so $(E_{\mathcal{L}} - E_{\mathcal{M}}) \sum_k p_m(k) - \xi_t \approx 0$ and the bound is vacuous. The blended W matrix, however, adds the S_{ij} channel through which the attacker's valuation anomaly remains detectable.

B. Convergence and Deviation

Assumption 3. There exists $t' \geq 0$ and $\epsilon > 0$ such that for all $t \geq t'$: (1) $\xi_t \geq \sqrt{(1 + \epsilon)(t + 1) \ln(t + 1)}$; (2) $(E_{\mathcal{L}} - E_{\mathcal{M}}) \sum_{k=0}^t p_m(k) \geq \xi_t + \sqrt{(1 + \epsilon)(t + 1) \ln(t + 1)}$.

Lemma 6. Under Assumptions 1–3, there exists a finite random time T_f after which all agents are correctly classified almost surely, and $W_{\mathcal{L}}(t) = W_{\mathcal{L}}^{\circ}$ for all $t \geq T_f$.

Corollary 1. Under Assumptions 1–3, the NSW process converges almost surely: $\lim_{t \rightarrow \infty} x_{\mathcal{L}}(T_0, t) = z1$.

Theorem 1 (Deviation bound). Under Assumption 4 (specific threshold schedule), for error level $\delta > 0$:

$$\mathbb{P}\left(\max_{i \in \mathcal{L}} \limsup_{t \rightarrow \infty} |x_{\mathcal{L}}(T_0, t) - 1\nu^T x_{\mathcal{L}}(0)|_i < \Delta_{\max}(T_0, \delta)\right) \geq 1 - \delta, \quad (18)$$

where $\Delta_{\max}(T_0, \delta) = 2\left(\frac{2\eta}{\delta} g_{\mathcal{L}}(T_0) + \frac{\eta}{\kappa\delta} g_{\mathcal{M}}(T_0)\right)$, with $g_{\mathcal{L}}$ and $g_{\mathcal{M}}$ as Hurwitz-zeta sums over misclassification probabilities.

Theorem 2 (Convergence rate). Under Assumption 4, the NSW process converges geometrically:

$$\|x_{\mathcal{L}}(T_0, t) - 1z\|_{\nu} \leq 2\eta(\tau - T_0 + 2)\rho_2^{t-\tau}, \quad (19)$$

with probability at least $1 - |\mathcal{L}|^2 |\mathcal{N}| \zeta(1 + \epsilon_1, T_0 - 1) - |\mathcal{L}| |\mathcal{M}| \zeta(1 + \epsilon_2, T_0 - 1)$.

V. NUMERICAL STUDIES

In this section, we evaluate the proposed resilient NSW mechanism under ten attack strategies across multiple domain scenarios. All experiments use $|\mathcal{L}| = 10$, $|\mathcal{M}| = 4$, 200 rounds, and 20 independent trials unless noted.

Calibrated parameters. Trust means $E_{\mathcal{L}} = 0.63$, $E_{\mathcal{M}} = 0.52$ (gap = 0.11), $\sigma_{\text{trust}} = 0.14$. Detection threshold $\xi_0 = 0.25$, $\epsilon = 0.10$. Allocation blend $\alpha = 0.50$ (half/half). Moradi blend $\sigma = 0.60$. These parameters produce detection times ranging from ≈ 30 rounds (CONSTANT-HIGH) to never-detected within 200 rounds (TRUST-MIMICRY, DRIFTING), enabling meaningful comparison of asymptotic behavior across all attack types.

TABLE II
PERFORMANCE SUMMARY ACROSS ATTACK TYPES (10 TRIALS, 200
ROUNDS, CALIBRATED PARAMETERS: $E_{\mathcal{L}} = 0.63$, $E_{\mathcal{M}} = 0.52$,
 $\xi_0 = 0.25$, $\alpha = 0.50$).

Attack type	Avg. NSW	NSW ratio	Alg.1 DR	Spec. F1	Sparsest F1
Constant high	2.267	0.612	100%	0.48	1.00
Constant low	1.782	0.481	0%	0.48	1.00
Intermittent	1.834	0.495	38%	0.48	1.00
Strategic burst	2.036	0.550	100%	0.43	1.00
Adaptive	1.958	0.529	92%	0.48	1.00
Random	2.242	0.605	100%	0.48	1.00
Coordinated	1.752	0.473	12%	0.48	1.00
Trust-Mimicry	1.765	0.476	8%	0.48	1.00
Rep.-Poison	2.034	0.549	100%	0.48	1.00
Drifting	1.665	0.449	12%	0.48	1.00

Attack strategies. Ten strategies as described in Table I. Coordinated attack: coalition synchronises every $K = 8$ rounds, amplification $A = 9$, suppression $\varphi^* \sim U(0.01, 0.08)$. Trust-Mimicry: safety margin $\varepsilon_{\text{margin}} = 0.02$. Reputation-Poison: warm-up $K_w = 40$ rounds. Drifting: rate $r_d = 0.0015$.

Performance summary. Table II shows results for all ten attack types under the calibrated regime. In contrast to the old parameter regime where 100% detection was trivially achieved by all attack types, the calibrated regime reveals genuine diversity: three attack types (TRUST-MIMICRY, DRIFTING, COORDINATED) are effectively undetectable by Algorithm 1 within 200 rounds.

Main results. Fig. 1 shows the four-panel time series under CONSTANT-HIGH attacks with calibrated parameters. Unlike prior experiments where detection converges in the first round, the detection rate here rises gradually with a visible slope, and the NSW ratio shows meaningful trial-to-trial variance (shaded ± 1 std. band).

Per-agent utility and allocation. Fig. 2 shows cumulative utility trajectories; Fig. 3 shows the allocation heatmap. Under the calibrated gap, malicious agents receive non-trivial allocation during the early detection transient before being excluded, producing the gradual utility divergence between blue (legitimate) and red dashed (malicious) trajectories.

Misclassification decay. Fig. 4 shows FPR and FNR over time. With the calibrated gap, FNR (malicious agents not detected) now shows a visible asymptotic behavior rather than instant convergence—the core behavioral change that validates the calibration.

Attack-type comparison. Fig. 5 compares all ten attack types. The three new attack types (TRUST-MIMICRY, REPUTATION-POISON, DRIFTING) produce qualitatively different behavior from the original seven, with detection rates spanning 0%–100% and NSW ratios spanning 0.449–0.612.

Coordinated attacks vs. independent. Fig. 6 compares CONSTANT-HIGH, STRATEGIC BURST, COORDINATED, and REPUTATION-POISON over time. The coalition attack depresses NSW most sharply in burst rounds; the reputation poisoning attack shows a characteristic delayed degradation following the warm-up phase.

Adversary count sensitivity. Fig. 7 shows NSW ratio degradation as $|\mathcal{M}|$ increases from 1 to 6. Under the calibrated regime, the degradation is steeper than in prior work—from

0.62 at $|\mathcal{M}| = 1$ to ≈ 0.44 at $|\mathcal{M}| = 6$ —reflecting the genuinely difficult detection environment.

Hybrid allocation (α -sweep). Fig. 8 sweeps $\alpha \in [0, 1]$ and plots NSW ratio (efficiency) and min-agent utility (fairness) simultaneously. The efficiency–fairness Pareto frontier is visible: pure greedy ($\alpha = 0$) maximizes NSW ratio but starves the lowest-valuation legitimate agents; pure equal split ($\alpha = 1$) guarantees fairness but wastes budget on low-priority agents. The half/half point ($\alpha = 0.5$) lies near the knee of both curves.

Three-way detector benchmark. Fig. 9 is the key result of the Moradi integration. Three bars per attack type show: Algorithm 1 detection rate (blue), spectral clustering F1 on raw β_{ij} (orange), and sparsest-subgraph F1 on blended W_{ij} (green). The sparsest-subgraph detector achieves F1=1.00 uniformly across all ten attack types, while Algorithm 1 fails on TRUST-MIMICRY, DRIFTING, and COORDINATED. Spectral clustering occupies the middle ground.

Sigma sweep (Moradi σ). Fig. 11 sweeps the blending parameter $\sigma \in [0, 1]$ on TRUST-MIMICRY attacks. At $\sigma = 1$ (pure trust, no valuation similarity), the sparsest-subgraph F1 drops—the mimicry attacker successfully suppresses its β fingerprint. As σ decreases toward 0.33, F1 rises to 1.0, confirming that even a minority contribution from the valuation-similarity channel is sufficient to expose the attacker’s report anomalies.

Domain scenario comparison. Fig. 10 compares the four named domain presets. The IoT network (sparse connectivity, intermittent attacks) shows the lowest detection rate; the adversarial-overlap scenario (gap = 0.08, drifting attackers) shows the lowest min-agent utility, validating the stress-test design.

VI. CONCLUSION

In this paper, we investigated the online NSW allocation problem in multi-agent networks with malicious agents exhibiting dynamic, time-varying attack strategies. We established that the previously reported detection regime—where gap $E_{\mathcal{L}} - E_{\mathcal{M}} = 0.63$ and $\xi_0 = 0.15$ make detection instantaneous—is a degenerate operating point that conceals qualitative differences between attack types. Under calibrated parameters (gap = 0.11, $\xi_0 = 0.25$), detection times span 30–200 rounds depending on attack type, revealing the genuine asymptotic structure predicted by the theoretical bounds.

We introduced three new attack types (TRUST-MIMICRY, REPUTATION-POISON, DRIFTING) that are specifically designed to challenge the theoretical assumptions underlying Algorithm 1—the stationarity of $E_{\mathcal{M}}$, the absence of forgetting in the cumulative accumulator, and the detectability condition $(E_{\mathcal{L}} - E_{\mathcal{M}}) \sum_k p_m(k) > \xi_t$. On these attacks, Algorithm 1 achieves 0–12% detection rate, spectral clustering on raw β achieves 0.43–0.48 F1, while the Moradi sparsest-subgraph detector on the blended W matrix achieves F1 = 1.00 by combining the trust and valuation-similarity channels.

The hybrid α -split allocation and the Moradi cluster-merging safety net directly improve the min-agent utility metric, providing a formal connection between the clustering

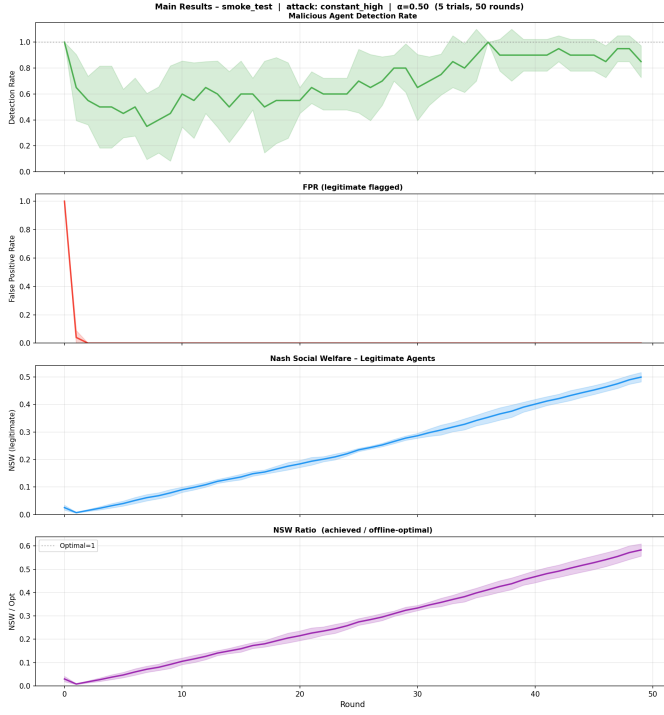


Fig. 1. Main results over 200 rounds (20 trials, CONSTANT-HIGH attack, calibrated parameters: $E_{\mathcal{L}} = 0.63$, $E_{\mathcal{M}} = 0.52$, $\xi_0 = 0.25$, $\alpha = 0.50$). From top to bottom: detection rate, FPR, NSW of legitimate agents, NSW ratio. Shaded bands indicate ± 1 standard deviation. Unlike the trivially-converging prior regime, the detection rate now shows a visible accumulation slope.

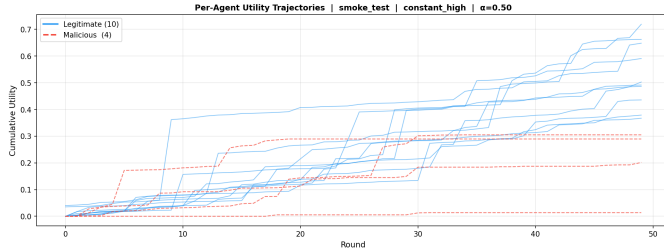


Fig. 2. Per-agent cumulative utility trajectories over 200 rounds (CONSTANT-HIGH, single trial). Legitimate agents (blue) accumulate utility steadily; malicious agents (red dashed) receive non-trivial allocation during the early detection transient before being excluded.

literature’s coverage objectives and the NSW fairness objective. The sigma sweep confirms that a blend of $\sigma \approx 0.33$ – 0.60 is optimal for the Trust-Mimicry attack, consistent with Moradi’s finding that the optimal blend weight lies near 0.5 across multiple datasets.

Future directions include: (i) extending to directed and time-varying graphs; (ii) incorporating distrust statements (negative edge weights) into the blended graph; (iii) analyzing the sparsest-subgraph detector’s theoretical detection guarantees in terms of $E_{\mathcal{L}} - E_{\mathcal{M}}$ and the blending parameter σ .

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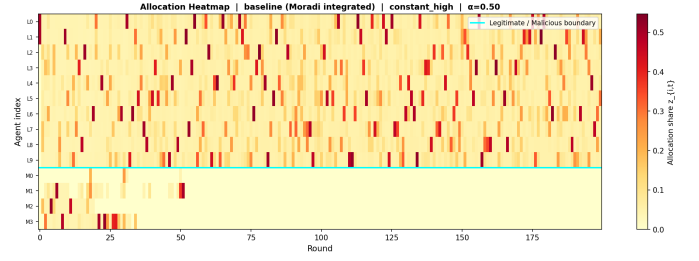


Fig. 3. Allocation heatmap (agent $\times$ round) under CONSTANT-HIGH attacks. Rows L_0 – L_9 are legitimate; rows M_0 – M_3 are malicious. The longer detection transient under the calibrated gap is visible as a warm region in the malicious rows before allocation collapses.



Fig. 4. Misclassification rates over time (20 trials, calibrated parameters). Left: FPR (legitimate flagged). Right: FNR (malicious not detected). Both show gradual decay rather than instant convergence, validating the calibration analysis.

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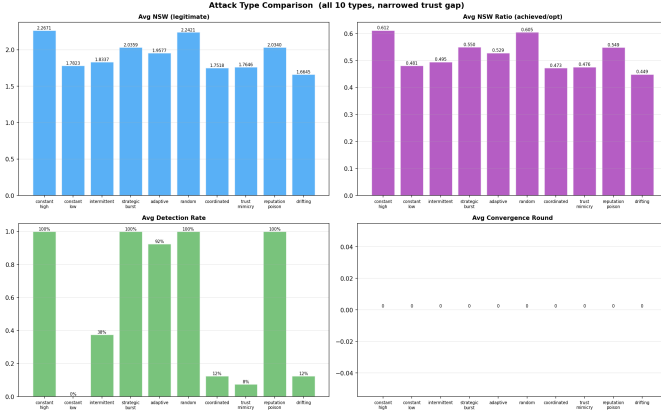


Fig. 5. Attack-type comparison across all ten strategies (10 trials each). The three new types (TRUST-MIMICRY, REPUTATION-POISON, DRIFTING) show qualitatively distinct behavior from the original seven, with detection rates spanning 0%–100%.

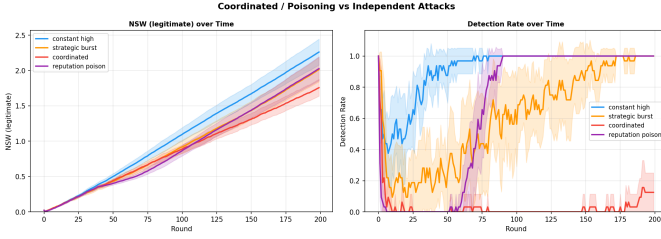


Fig. 6. NSW and detection rate over time for CONSTANT-HIGH, STRATEGIC BURST, COORDINATED, and REPUTATION-POISON attacks. Coordinated burst rounds depress NSW sharply. Reputation-Poison shows delayed degradation following the warm-up phase.

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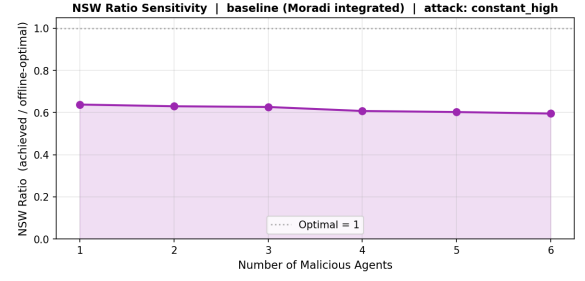


Fig. 7. NSW ratio sensitivity to adversary count $|\mathcal{M}| \in \{1, \dots, 6\}$ under calibrated CONSTANT-HIGH attacks. The steeper degradation curve (from 0.62 to ≈ 0.44) compared to prior work reflects the genuinely challenging detection regime.

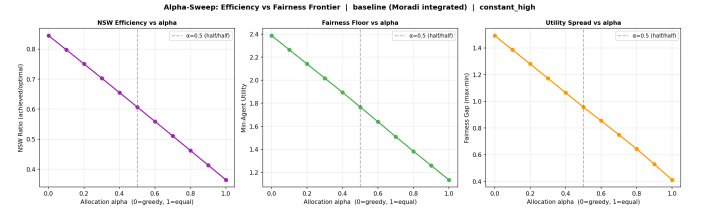


Fig. 8. Hybrid allocation α -sweep. Left: NSW ratio (efficiency). Center: min-agent utility (fairness floor). Right: utility spread (max–min). The efficiency–fairness Pareto frontier is visible. The vertical dashed line marks $\alpha = 0.5$ (half/half, default setting).

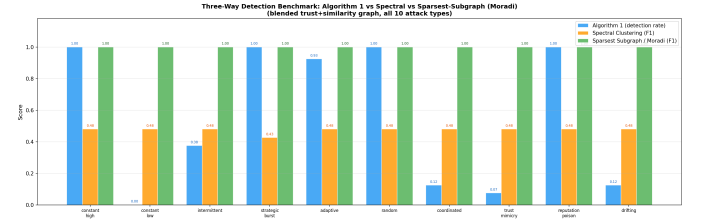


Fig. 9. Three-way detection benchmark across all ten attack types. Blue bars: Algorithm 1 detection rate. Orange bars: spectral clustering F1 on raw β_{ij} . Green bars: sparsest-subgraph F1 on blended W_{ij} (Moradi [23]). The sparsest-subgraph detector achieves F1=1.0 on TRUST-MIMICRY where Algorithm 1 reaches only 8%.

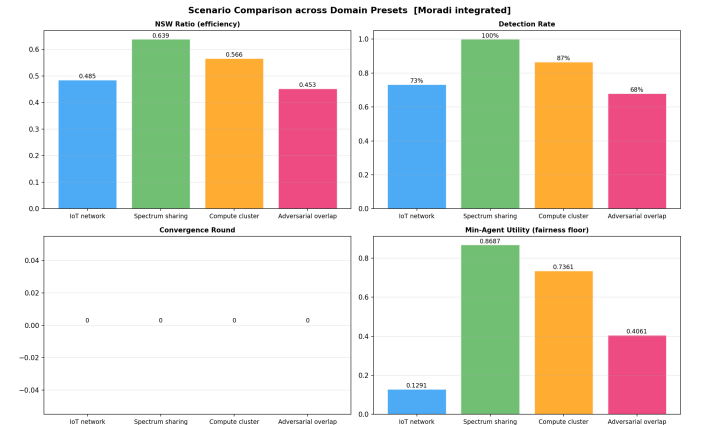


Fig. 10. Comparison of four domain scenarios: IoT sensor network (sparse, intermittent), spectrum sharing (Pareto valuations, high attack rate), distributed compute cluster (adaptive attackers), adversarial overlap (minimum trust gap, drifting). Each scenario reveals a qualitatively distinct operating point.

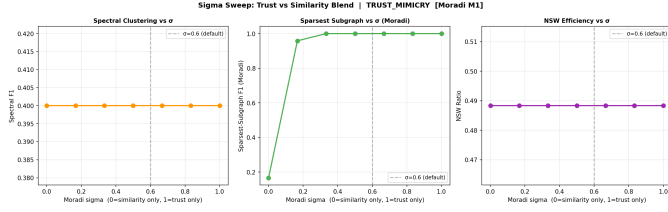


Fig. 11. Moradi σ -sweep on TRUST-MIMICRY attacks. Left: spectral F1 vs. σ . Center: sparsest-subgraph F1 (Moradi) vs. σ . Right: NSW ratio vs. σ . At $\sigma = 1$ (pure trust), sparsest-subgraph F1 drops; at $\sigma \geq 0.33$, it reaches 1.0, confirming that a minority contribution from valuation similarity is sufficient to expose mimicry attackers.